

# ARIANA CHIN

University of California

520 Portola Plaza

Los Angeles, CA 90095

arianagchin@math.ucla.edu

---

**RESEARCH INTERESTS:** Zamolodchikov periodicity, cluster algebras, algebraic combinatorics

**ADVISOR:** Pavel Galashin

## EDUCATION

### University of California, Los Angeles

*PhD in Mathematics*

Expected June 2027

*PhD Candidate*

May 2025

*M.A. in Mathematics*

March 2024

*Dissertation Year Award* - UCLA Department of Mathematics fellowship

2026-2027

*Liggett Teaching Award* - awarded to a teaching assistant for outstanding performance

2025-2026

*Balbes Award* - awarded to the top performing female graduate student

2023-2024

*Teaching Experience:*

- Upper Division: Probability and Statistics, Enumerative Combinatorics, Real Analysis
- Lower Division: Discrete Mathematics, Linear Algebra, Multivariable Calculus

### University of California, Berkeley

*B.A. in Mathematics, B.A. in Computer Science*

May 2022

*Magna Cum Laude*

GPA: 3.96

- Percy Lionel Davis Award - awarded to a senior or seniors who have demonstrated excellent scholarship in mathematics
- Regents' and Chancellor's Scholar

Languages: Python, Java, Scheme, C, JavaScript, RISC-V, Matlab, LaTeX, SQL, MongoDB, DynamoDB

## RESEARCH

“Half-periodicity of Zamolodchikov periodic cluster algebras” [[pdf](#)][[arXiv: 2602.15140](#)] 2026

- Proved the bipartite belt of length  $h + h'$  is a maximal green sequence for every Zamolodchikov periodic B-matrix
- Proved for any Zamolodchikov periodic B-matrix, the form at the half-period is a permutation of the cluster variables of order at most two

“Classification of Zamolodchikov periodic cluster algebras” [[pdf](#)][[arXiv: 2510.18031](#)] 2025

- Proved that the Zamolodchikov periodic cluster algebras are in bijection with pairs of commuting (not necessarily reduced or simply-laced) Cartan matrices of finite type
- Classified all such pairs into 29 infinite families and 14 exceptional types in addition to the 6 infinite families and 11 exceptional types in Stembridge's classification
- Showed these pairs are in 1-to-4 correspondence with nonnegative W-cells for products of two dihedral groups, which encode the action of the standard generators on the Kazhdan–Lusztig basis of the associated Iwahori–Hecke algebra

University of Minnesota, Twin Cities - Research Experience for Undergraduates (REU)

2021

*Algebra, Combinatorics, Representation Theory*

Minneapolis, MN

“F-Polynomial Ratios in the  $r$ -Kronecker” [[pdf](#)][[slides](#)]

- Joint with Noah Caplinger, Nyah Davis, and Swapnil Garg
- Explored limiting cluster variables and F-polynomial ratios in the Kronecker cluster algebra
- Presented an alternate proof for the formal power series expansion of the limit in the 2-Kronecker case, and a closed form in the general r-Kronecker case

“Lattice Models and Puzzles for Dual Weak Symmetric Grothendieck Polynomials” [[pdf](#)][[slides](#)]

- Joint with Elisabeth Bullock, Noah Caplinger, Nyah Davis, and Gahl Shemy
- Constructed a lattice model for the dual weak symmetric Grothendieck polynomials, with a negative result for stable symmetric Grothendieck polynomials
- Presented boundary conditions on a lattice model for Littlewood Richardson coefficients for dual weak symmetric Grothendieck polynomials

Ref: <https://www-users.cse.umn.edu/~reiner/REU/REU.html>

Boise State University - Complexity Across Disciplines REU

2019

*Number Theory, Quantum Cryptography*

Boise, ID

“The Polynomial Learning With Errors Problem and the Smearing Condition” [[paper](#)][[arXiv:2008.04459](#)]

- Joint with Liljana Babinkostova, Aaron Kirtland, Esther Plotnick, and Vladyslav Nazarchuk
- *Journal of Mathematical Cryptology* (JMC), 2022

“The Ring Learning With Errors Problem: Spectral Distortion” [[pdf](#)][[arXiv:2007.13189](#)]

- Joint with Liljana Babinkostova, Aaron Kirtland, Esther Plotnick, and Vladyslav Nazarchuk
- Accepted for publication by *Involve* journal of mathematics as of 7/2022
- Investigated the security of the Learning with Errors (LWE) problem, answering questions regarding smearing and spectral distortion used as conditions for a successful attack
- Presented a smearing-based attack and a closed form for the spectral distortion of cyclotomic polynomials

## CONFERENCES, WORKSHOPS & TALKS

- AMS Sectional: Diagrammatics for Cluster Algebras and Physics - Speaker  
*Arizona State University* November 2026  
Tempe, AZ
- University of Michigan Combinatorics Seminar - Invited Speaker  
*University of Michigan, Ann Arbor* October 2026  
Ann Arbor, MI
- Richard P. Stanley Seminar in Combinatorics - Invited Speaker  
*Harvard University* October 2026  
Boston, MA
- IPAM Tutorials: Quantum Topology, Character Varieties - Lecturer  
*Institute for Pure and Applied Mathematics, UCLA* September 2026  
Los Angeles, CA
- Formal Power Series and Algebraic Combinatorics (FPSAC) 2026 - Presenter  
Seattle, WA
- CMND 2026: Algebraic Combinatorics and Applications - Lightning talk  
South Bend, IN
- Cornell Combinatorics Seminar (2026) - Invited Speaker  
*Cornell University* April 2026  
Ithaca, NY
- UIUC Combinatorics Seminar (2026) - Invited Speaker  
*University of Illinois, Urbana-Champaign* April 2026  
Urbana-Champaign, IL
- ALGECOM 2026 - Presenter  
West Lafayette, IN
- UMN Combinatorics Seminar (2026) - Invited Speaker  
*University of Minnesota, Twin Cities* April 2026  
Minneapolis, MN
- NCUWM 2026 - Invited Panelist  
Lincoln, NE

- |                                                                     |                 |
|---------------------------------------------------------------------|-----------------|
| • Summer School on Algebraic Combinatorics (SSAC) 2025              | Seoul, Korea    |
| • Formal Power Series and Algebraic Combinatorics (FPSAC) 2025      | Sapporo, Japan  |
| • Graduate Student Combinatorics Conference (GSCC) 2025 - Presenter | Los Angeles, CA |
| • Formal Power Series and Algebraic Combinatorics (FPSAC) 2024      | Bochum, Germany |

## OTHER INTERESTS

|                                           |           |
|-------------------------------------------|-----------|
| UCLA Women in Mathematics - organizer     | 2024-2026 |
| UCLA Symphony, UCLA Philharmonia - violin | 2022-2026 |
| UC Berkeley Symphony Orchestra - violin   | 2019-2022 |
| UC Berkeley Chamber Orchestra - violin    | 2018-2019 |